Total No. of Questions : 6]

SEAT No. :

P369

[Total No. of Pages : 2

BE/Insem/APR-17
B.E. (Mechanical)
POWER PLANT ENGINEERING
(2012 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.*
2) *Draw neat diagrams wherever necessary.*
3) *Use of scientific calculator is allowed.*
4) *Use of steam table and mollier chart is allowed.*
5) *Assume suitable data wherever necessary.*
6) *Figures to the right indicate full marks.*

CEGP013091 49.248.216.238 11/03/2019 13:49:21 static-238

Q1) a) What is Carbon Credits? What is its significance in current energy scenario? **[4]**

b) For a thermal power plant of 210 MW, following data is available : **[6]**

Capital cost is Rs. 10,000 per kW. The fixed cost is 13% of the investment cost. At full load variable cost is 1.3 times the fixed cost per year. Assuming the variable cost is proportional to the energy produced, find the generating cost of the when the plant is running at full capacity condition.

OR

Q2) a) Maximum load on the power plant is 60 MW. The load having demand of 30 MW, 20 MW, 10 MW and 14 MW are connected to the plant. The capacity of the plant is 80 MW and annual load factor is 0.5 Estimate **[6]**

i) Average load on the power plant.

ii) Energy supplied per year.

iii) Demand factor.

iv) Diversity factor.

b) Elaborate the different parameters considered for calculating the cost of electric energy. **[4]**

P.T.O.

Q3) a) Steam at a pressure of 15 bar and 250ºC is expanded through a turbine at first to a pressure of 4 bar. It is then reheated at constant pressure to the initial temperature of 250ºC and is finally expanded to 0.1 bar. Draw the T-S diagram and estimate the work done per kg of steam flowing through the turbine and amount of heat supplied during the process of reheat. Compare the work output when the expansion is direct from 15 bar to 0.1 bar without any reheat.

Assume all expansion processes to be isentropic. **[6]**

b) Enlist the sources of air and effects of air leakage in the condenser on the performance of the condenser. **[4]**

OR

Q4) a) With the help of neat sketch explain pressurized fluidized bed combustion boiler. **[5]**

b) With the help of neat sketch explain regenerative reheat cycle for thermal power plant. **[5]**

Q5) a) Write in detail Nuclear Waste Disposal. **[6]**

b) Enlist the advantages and disadvantages of Hydro Electric Power Plant (HEPP). **[4]**

OR

Q6) a) With the help of neat sketch explain Boiling Water Nuclear Reactor. **[5]**

b) Explain in detail the parameters considered for selection of site for Hydro Electric Power Plant (HEPP). **[5]**

CEGP013091 49.248.216.238 11/03/2019 13:49:21 static-238